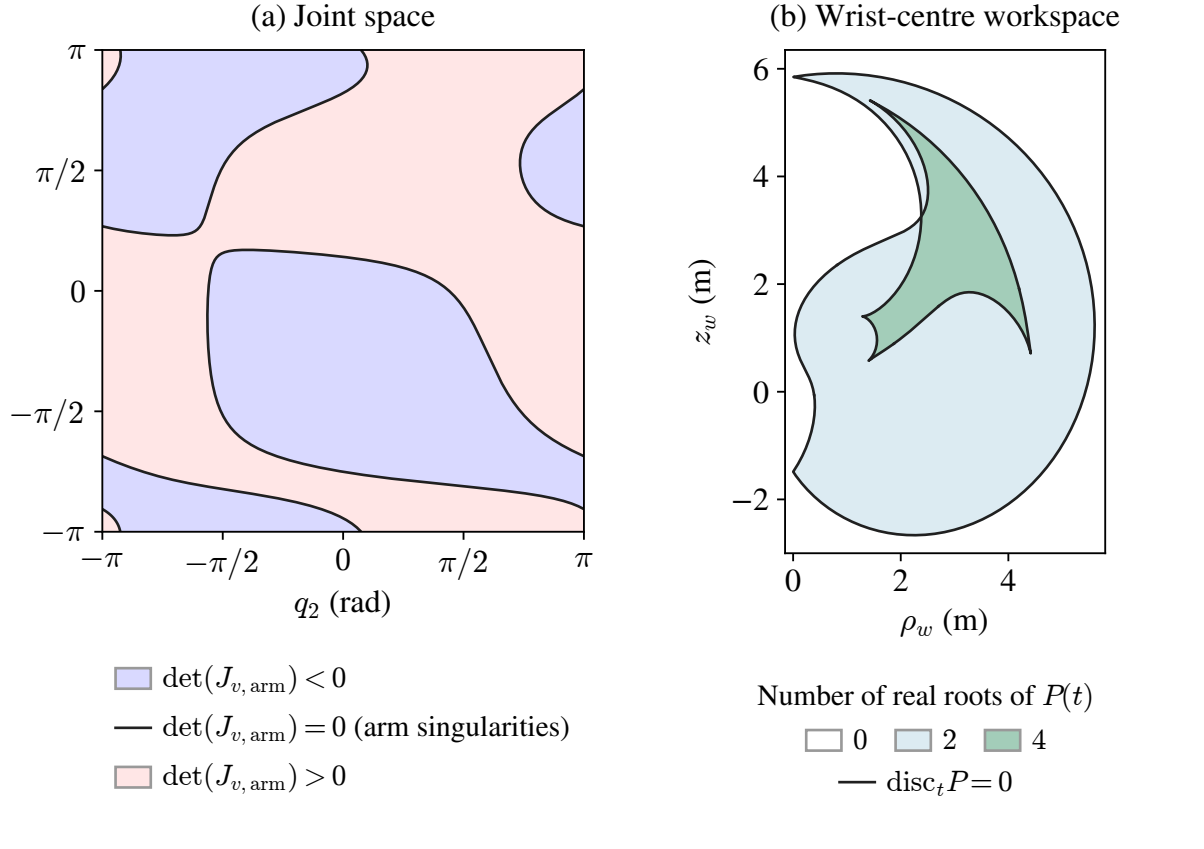

(a) Joint space
(b) Wrist-centre workspace
π
$\pi/2$
0
$-\pi/2$
$-\pi$
q_2 (rad)
6
4
2
0
−2
z_w (m)
ρ_w (m)
$\det(J_{v,\mathrm{arm}}) < 0$
$\det(J_{v,\mathrm{arm}}) = 0$ (arm singularities)
$\det(J_{v,\mathrm{arm}}) > 0$
Number of real roots of $P(t)$
0
2
4
$\mathrm{disc}_t P = 0$